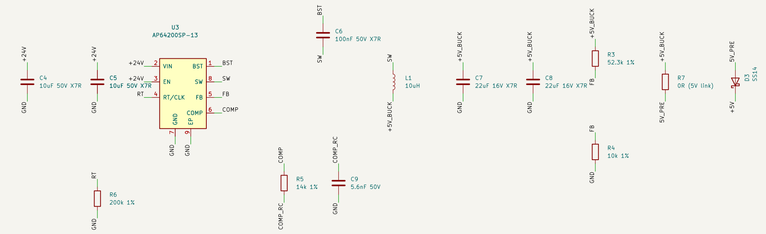
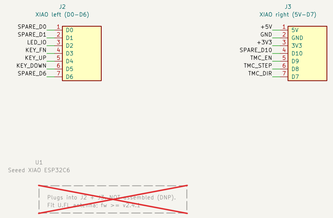
## 24 V INPUT: fuse -> reverse-polarity P-FET -> TVS / bulk. PSUs at nominal 24 V (SMBJ24A): do not trim up.



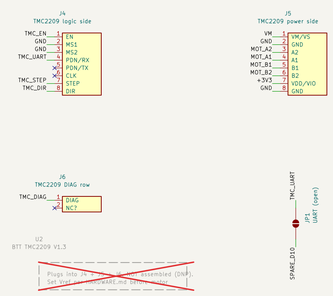

## 24 V -> 5 V BUCK (AP64200, Table 1 ● 500 kHz). R7 / R2 are 0R links for per-section idle-current measurement.



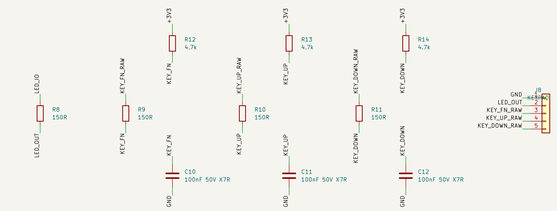

## XIAO ESP32C6 SOCKETS (pin map v2.4.x, unchanged)



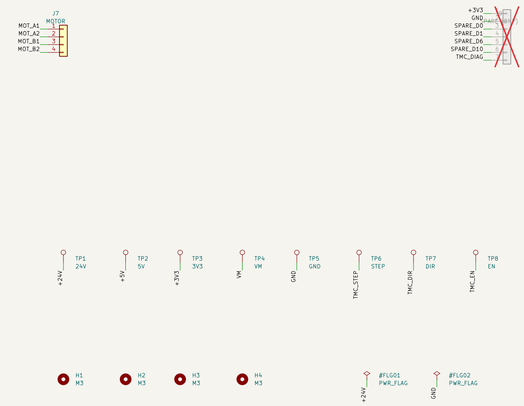

## TMC2209 V1.3 SOCKETS: MS1/MS2 = GND -> 1/8 step. VIO = 3V3. JP1 open by default.




## KEYPAD (5-pin JST-XH: GND, LED, Fn, Up, Down): 150R series, 4.7k pull-up, 100nF at the MCU end




## MOTOR / SPARE (J9 not fitted) / TEST POINTS / MOUNTING




## J6 DIAG-row pin order to be confirmed by 'Measure the physical XIAO and TMC2209 modules'.


Map: github.com/linosteenkamp/DFR-RollerBlinds/issues/1
XIAO ESP32C6 + BTT TMC2209 V1.3 carrier, 24 V in

Sheet: /
File: carrier-board.kicad_sch

**Title: DFR-RollerBlinds carrier board**

Size: A1 | Date: 2026-11-06 | **Rev: A**
KiCad E.D.A. 10.0.8 | Id: 1/1